

Exercise Sheet 8 - Linear Programming

Deadline for submission: Wednesday, 10 December 2025 at 8:00.

Discussed in exercise session: Friday, 12 December 2025 at 10:15.

Exercise 8.1 - Simplex Algorithm: Maximization

Consider the following **maximization** problem:

$$\text{max} \quad Z = 4X + 6Y$$

$$\begin{aligned} \text{subject to} \quad & -X + Y \leq 11 \\ & X + Y \leq 27 \\ & 2X + 5Y \leq 90 \\ & X, Y \geq 0 \end{aligned}$$

1. Find the maximum value Z .
2. At what values of X and Y does the maximum occur?
3. State the basis variables in the maximum.
4. Draw on paper the linear constraints, mark the feasible region and indicate all basic feasible solutions (BFS) as well as the basis variables for each BFS.

Exercise 8.2 - Simplex Algorithm: Minimization

Consider the following **minimization** problem:

$$\text{min} \quad Z = 3A + 2B$$

$$\begin{aligned} \text{subject to} \quad & 2A + B \geq 6 \\ & A + B \geq 4 \\ & A, B \geq 0 \end{aligned}$$

1. Find the minimum value Z .
2. At what values of A and B does the minimum occur?
3. State the basis variables in the minimum.
4. Draw on paper the linear constraints, mark the feasible region and indicate all basic feasible solutions (BFS) as well as the basis variables for each BFS.

Exercise 8.3 - Simplex Algorithm: Minimization

Consider the following **minimization** problem:

$$\begin{aligned} \text{min} \quad & Z = 11A + 8B \\ \text{subject to} \quad & 2A + B \geq 12 \\ & A + 2B \geq 12 \\ & A + B \geq 10 \\ & 3A + 4B \leq 60 \\ & A, B \geq 0 \end{aligned}$$

1. Find the minimum value Z .
2. At what values of A and B does the minimum occur?
3. State the basis variables in the minimum.
4. Draw on paper the linear constraints, mark the feasible region and indicate all basic feasible solutions (BFS) as well as the basis variables for each BFS.